Deformable Electrostatic Linear Motor

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Abstract— **This paper describes high-power electrostatic linear motor that is *deformable*. The motor consists of pairs of flexbiel printed circuit films. It is characterized by superior output performance; typically, the motor can generate several tens of newtons, or slider speed can exceed 1m/s. In this paper, we focus on its flexibility. Due to the flexiblity of the films used, the motor can be mechanically deformed. On top of that, the motor can even operate while being deformed. Our analyses show that the motor can operate even when it is bent 180 degrees. In this paper, thrust force characteristics of the motor under deformed state are summerized and some examples of motor deformation are shown.**

Keywords— **Electrostatic Motor, Linear Motor, Actuator, Flexible Printed Circuit, Film Actuator**

IP Info.— **This study was carried out based on the patented technology, United States Patent ♯5,448,124**

I. Introduction

Recently there have been great interests on human-friendly robotic technolgoies. One of the key issues in the technologies would be realization of soft mechanical systems that can gently interact with humans. As the actuators for such soft mechanical systems, many kinds of soft or deformable actuators have been proposed, including pnueumatic actuators, polymer actuators, and SMA actuators.

In this paper, we introduce another type of deformable actuator. The actuator is a linear synchronous motor driven by electrostatic force. It is called *Dual Excitation Multiphase Electrostatic Drive* (DEMED), which was already proven to be high-power[1] and highly-controllable[2]; typically, it can generate thrust force of several tens of newtons and slider speed can exceed 1m/s. In this paper, we focus on its deformability and show that the motor can be deformed even during operations.

II. Basic Working Principle [1]

Fig.1 shows the basic structure of the electrostatic motor. The motor consists of two plastic films, both of which contain three-phase parallel electrodes, typically aligned with regular intervals of 200μm. These two films function as a stator and a slider, respectively. In the gap between two films, dielectric liquid (eg. Fluorinert (3M) or silicon oil) is injected to enhance dielectric breakdown strength at the gap. Also, glass beads with diameters of around 20μm are scattered in the gap to reduce friction force.

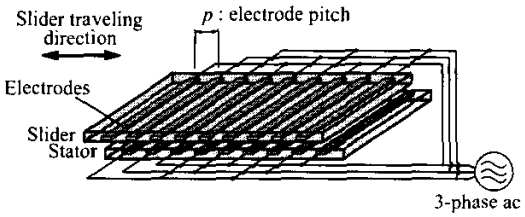


Fig. 1. Basic structure

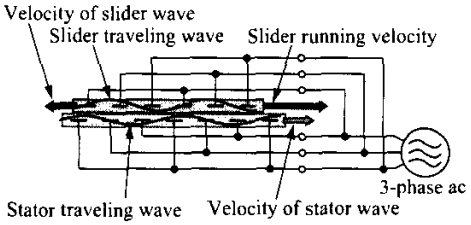


Fig. 2. Working principle

Fig. 2 shows the working principle. By applying three-phase ac voltage to both three-phase electrodes of the slider and stator, with reversed connecting orders to each other, sinusoidal voltage distributions running toward opposite directions are excited on the films. The electrostatic interaction between the distributions generate thrust force, which makes slider run in a speed proportinal to the applied frequency. Thrust force of the motor is proportinal to square amplitude of applied ac voltage. Typically, the required amplitude for motor operation is 1kV-2kV$_{0-p}$.

III. Deformable Motor

A. Thrust force in normal state

This motor has ripples in thrust force characteristics; the force generation capability changes depending on the slider displacement. Fig. 3 shows thrust force generation capability at each displacement, which is analyzed under the condition of constant voltage amplitude. Since the electrodes are connected in three-phase, the thrust force characteristics repeat every three electrode pitches. In this plot, we define F_{max} and F_{min} as the maximum and minimum thrust forces in the variation.

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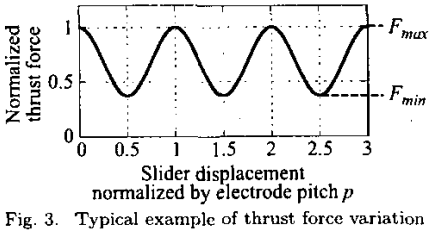


Fig. 3. Typical example of thrust force variation

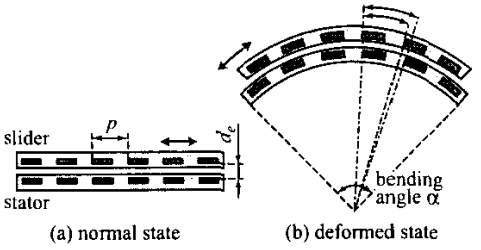


Fig. 4. Electrode misalignment when bent in traveling direction

B. Thrust force under simple deformation

When the motor is deformed, thrust force characteristics change depending on the deformation. For the simplest deformation, where the motor is bent in a constant curvature along traveling direction as in Fig. 4, thrust force characteristics were analyzed and experimentally verified in [3], of which results are shown in Fig. 5. In the analysis, friction force between slider and stator is ignored.

The major reason that thrust force characteristics change as the motor deforms is electrode misalignment occured by deformation. In normal state, electrodes in both stator and slider are aligned with the same intervals as in Fig. 4(a). However, when the motor is deformed as in Fig. 4(b), the electrode intervals along moving circle differ between slider and stator.

On the other hand, when the motor is bent along the transverse direction, the electrode misalignment does not occur. Therefore, provided that friction can be ignored, thrust force characteristics do not change even when the motor is bent. Fig. 6 shows the result of an experimental verification, in which the motor under different curvature pulled up weights. The maximum weights that the motor could pull up are plotted. The result shows that bendings in the transverse direction do not affect thrust force, for a certain range of bending curvature.

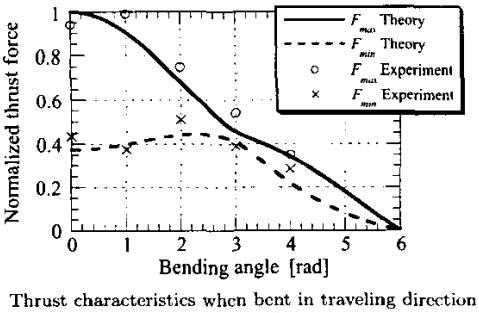


Fig. 5. Thrust characteristics when bent in traveling direction ($p = 200\mu$m, $d_e = 100\mu$m in Fig. 4.)

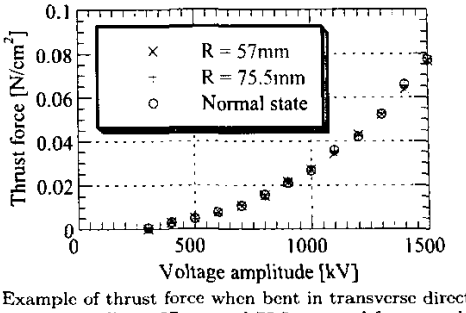


Fig. 6. Example of thrust force when bent in transverse direction for curvature radiuses 57mm and 75.5mm, and for normal state

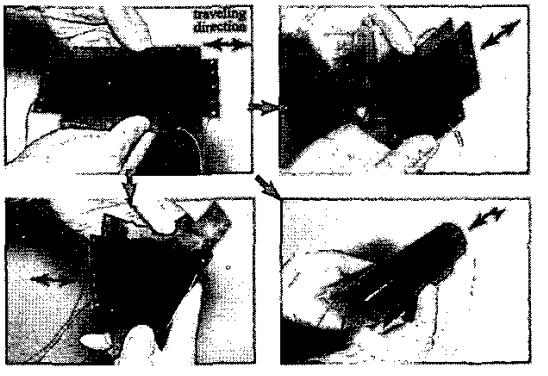


Fig. 7. Examples of deformations

C. Examples of Deformations

Fig. 7 shows some examples of deformations. On the upper left is a prototype motor in normal state. The prototype consists of two pairs of slider and stator films; the slider films are sandwiched between stator films. The overall dimensions are 120mm in width, 160mm in length. In the upper right in Fig. 7, the motor is bent along the traveling direction, lower right along the transverse direction, and lower left along a diagonal direction. Under all of these deformations, the motor was confirmed to be able to operate.

IV. Conclusion

In this paper, we introduced *deformable* electrostatic linear motor. The motor can be operated in various deformations. Some examples of motor deformation were shown.

Acknowledgement

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